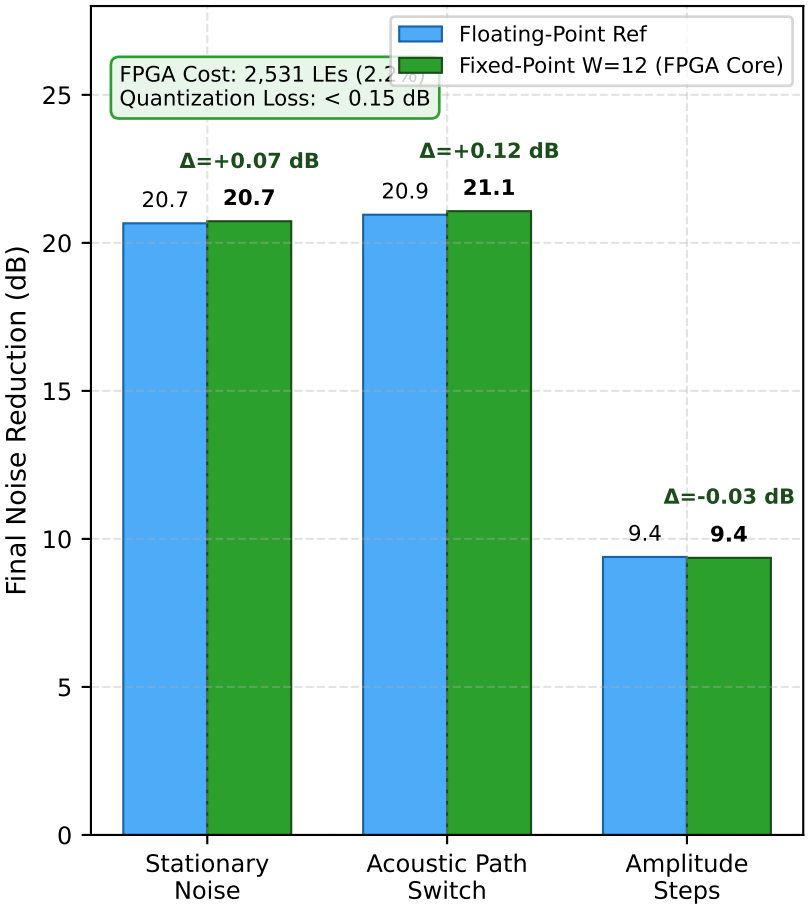

FPGA Implementation Status & Complexity Comparison
Cyclone IV E EP4CE115F29C7

NLMS-Shift (Power-of-2 Dividerless)
SYNTHESIZED (Quartus Prime 25.1)
• LEs: 2,531 (2.2%) • Registers: 1,113
• Multipliers: 8 (9-bit) • Divider: None

LMS (Fixed Step Size)
PENDING SYNTHESIS (Unsynthesized)
• Theoretical profile: 2L+1 mults, no divider
• Hardware cost pending Quartus run

Standard NLMS (Serial Restoring Divider)
PENDING SYNTHESIS (Unsynthesized)
• Theoretical profile: 2L+3 mults + serial divider
• Hardware cost pending Quartus run

Methodology Note: The previously reported 10.0 dB vs 20.7 dB comparison mixed a multi-scenario bitwidth average with stationary-only float data. Strictly matched evaluations show matched-scenario results with absolute noise-reduction differences below 0.15 dB for the three evaluated scenarios. A cross-algorithm hardware Pareto frontier remains deferred until LMS and NLMS synthesis runs complete.